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Design Rationale

The message of this visualization is that added sugar does not track processing uniformly; it concentrates inside a specific corner of the grocery store: sweet ultra-processed categories such as candy, cookies, cakes, cereal, snack bars, and ice cream. The naive narrative is "ultra-processed equals high sugar," but the data tells a more nuanced story: plenty of ultra-processed categories (pizza, packaged meat, cheese, prepared meals) sit at low-to-moderate sugar levels, while a handful of sweet categories push the mean far above everything else. I designed a single chart (rather than a two-panel layout) to let readers discover this pattern themselves: the red "sugar trap" quadrant in the upper-right reveals where risk actually lives, the teal minimally-processed zone anchors a baseline on the left, and the gray middle field shows that most ultra-processed foods are not, in fact, sugar bombs. The 76% callout in the corner, drawn from the predicted NOVA classification (FPro_class = 3) rather than the continuous FPro score, gives scale context: ultra-processed foods dominate grocery shelves, but only a sweet subset drives the sugar problem.

Following Munzner's channel-effectiveness ranking, I assign the two most important data dimensions (processing score FPro and mean sugar content) to **position on a common scale** on the x and y axes, the strongest magnitude channel for ordered attributes. Position on the x-axis encodes the continuous FPro (0 = unprocessed, 1 = ultra-processed), carrying far more resolution than the four discrete NOVA classes. **Shape**, a top-ranked identity channel, encodes the categorical processing group: triangles are ultra-processed (FPro > 0.75), circles moderately processed ($0.55 \leq \text{FPro} \leq 0.75$), and squares minimally processed (FPro < 0.55). This satisfies the *expressiveness* criterion, as shape distinguishes a nominal attribute without implying false order. Shape encoding also serves an explicit accessibility goal: viewers with red-green color vision deficiency (deuteranopia or protanopia) cannot reliably distinguish red from teal by hue alone, so the triangle, circle, and square marks let them identify the processing group without depending on color at all. **Color saturation** then encodes sugar intensity within each group: darker red signals higher sugar, lighter red signals lower sugar, so the reader immediately sees that sweet ultra-processed triangles are visually heavier than savory ones. This is an *effectiveness* advantage, since luminance and saturation are strong channels for ordered data. **Mark area** (2D size) scales with category item count raised to the 0.85 power, giving a ~15x area ratio between a 100-item and a 2,000-item category. A dashed vertical rule at FPro = 0.75 anchors the ultra-processed threshold, and two faint tinted zones reinforce the regions spatially. Dash pattern on the marginal KDE curves provides a further redundant encoding layer, ensuring no critical distinction depends on hue alone.

Before plotting I filtered to rows with non-null FPro and Sugars, and dropped 29 items reporting sugar > 100 g/100g (physically impossible, as with a belVita biscuit listed at 180 g/100g of sugar, clearly a data entry error). The remaining 26,221 items span Walmart, Target, and Whole Foods. From these items I computed category-level means of FPro and sugar, along with item counts, then kept only categories with at least 100 items for stable estimates, leaving 49 categories out of the original set. I cross-referenced FPro against FPro_class (NOVA) as a data quality check: every item with FPro > 0.75 is NOVA 3, confirming zero false positives in the ultra-processed zone; however, 25% of NOVA 3 items fall between 0.50 and 0.75, so the FPro > 0.75 threshold is conservative. The 76% callout therefore uses FPro_class = 3 rather than the FPro threshold to avoid understating ultra-processed prevalence. Because GroceryDB's raw category slugs are not human-readable, I mapped each to a clean label. Because "Cakes" and "Cookies and Biscuits" land at essentially identical positions, I merged them into a single bubble labeled "Cookies and Cakes" to prevent a label collision. All nutritional values are pre-normalized to g per 100 g of product.

My earlier drafts went through several revisions. A mean-sugar bar chart by NOVA class was too coarse; a two-panel density-plus-category-bar layout violated the one-visualization rule; an earlier scatter used color alone for a three-way split (red, teal, gray), which violated *expressiveness*: the reader could not tell whether non-sweet ultra-processed categories were hiding among the gray middle. The current version fixes that by encoding processing level with shape, so every ultra-processed category is visibly a triangle. The reader can verify directly that the lighter-red triangles (Pizza, Chips, Sauces, Packaged Meat, Prepared Meals, Sausage and Bacon, Jerky) sit at low sugar while the darker-red triangles (Candy and Pastry, Cookies and Cakes, Cereal, Snack Bars, Ice Cream, Pudding, Muffins) climb the y-axis. The category-level Pearson correlation between FPro and sugar is $r = 0.42$ (moderate, not strong), which is itself evidence for the split framing. Additional refinements include fading 34 unlabeled marks to background gray so the eye fixates on story-critical categories via pre-attentive contrast, using power-0.85 size scaling, and tightening the y-axis to 0 to 45 rather than wasting space above the data maximum of about 39. The main limitation is that item-level variance within a category is hidden inside each mark: a mostly low-sugar ultra-processed category still contains individual high-sugar items. I accepted this because the argument is about where sugar concentrates in the grocery landscape as a whole, not about individual products.